METRIC

A-A-50621
23 February 1993
SUPERSEDING
MIL-L-17080C(SHIPS)
16 September 1968

COMMERCIAL ITEM DESCRIPTION

LOUDSPEAKERS, PERMANENT MAGNET FOR SHIPBOARD ENTERTAINMENT SYSTEMS

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-L-17080C which is canceled.

Abstract. This Commercial Item Description covers requirements for permanent magnet loudspeakers suitable for use on naval ships.

Salient characteristics. Loudspeakers furnished under this Commercial Item Description shall be of the permanent magnet direct radiator type. The speaker enclosure, designed for bulkhead mounting, shall contain a loudspeaker unit with a cone having a nominal 203 mm diameter, a line matching transformer, a volume selector switch and a channel selector switch.

1. Integral components. Loudspeaker assemblies shall consist of the following integral components manufactured in accordance with the following specifications or as specified herein:

(a) Loudspeaker.

EIA-SE-103	Speakers for sound equipment
EIA-157	Method for determining air gap flux density and energy.
EIA-276	Acceptance Testing of Dynamic Loudspeakers

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Commander, Naval Sea Systems Command, SEA 03R42, 2531 Jefferson Davis Hwy., Arlington, VA 22242-5160.

AMSC N/A FSC 5965

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(b) Volume and Channel Selector Switch (Rotary).

EIA-520F000 Sectional Specification for Rotary Switches of Certified Quality (Low Current Capacity - 2 Amperes or Less)

(c) Enclosure. Bulkhead enclosures shall be as specified herein.

2. Enclosure.

(a) Loudspeaker enclosure. The loudspeaker enclosure shall have three 13 mm diameter holes in the back plate for bulkhead mounting. Mounting details shall be as shown on figure 1. Except for the increased depth incident to opening the hinged front cover 90 degrees and the clearance dimensions required to install, remove, or service the loudspeaker. The enclosure shall not be not greater than the following dimensions·

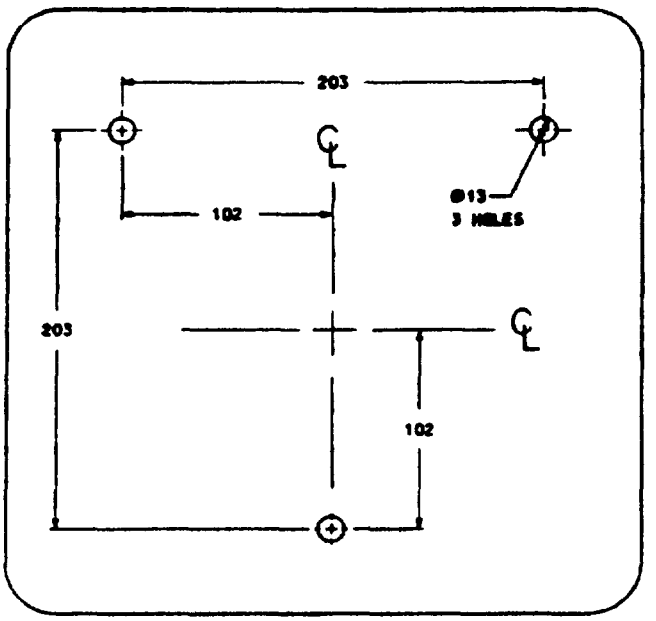
Width - 380 mm
Height - 380 mm
Depth - 230 mm

1. Enclosure openings. The enclosure shall have an opening for the loudspeaker face. This opening and any port openings required to meet the specified performance characteristics, shall be covered with metal mesh or perforated metal not less than 1 mm thick. Mesh openings or perforations shall be not larger than 2 mm.

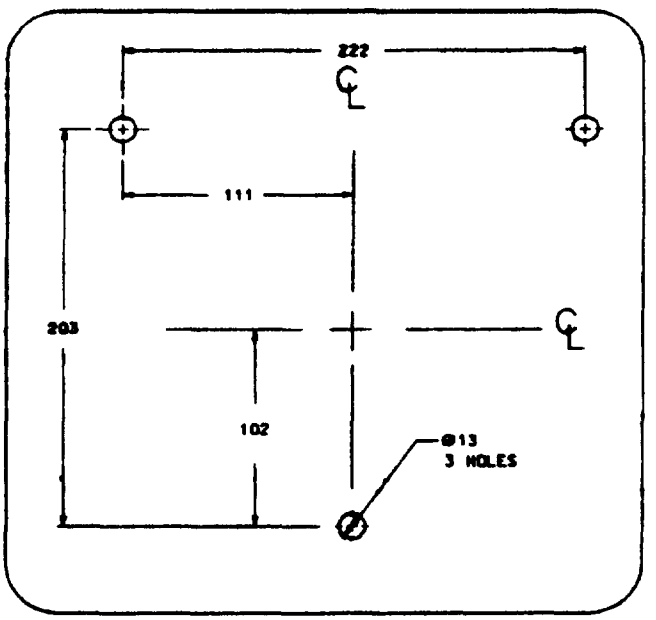
2. Enclosure construction. The enclosure shall be provided with a hinged front cover, held closed by a catches requiring no tools for operation. Opening the cover shall provide ready access to all interior components and to the enclosure mounting holes. Opening the enclosure shall not involve the disturbance of incoming cables or stuffing tubes. Component mounting fasteners shall be secured so as to not require demounting of the enclosure for component replacement. Components shall be mounted so that their terminal markings are easily visible with the enclosure door open. Damping material, if used, shall be fastened positively.

3. Cable entrance. Facilities shall be provided to admit cables through the bottom of the enclosure utilizing stuffing tubes.

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TYPE LS-443/WD-1



TYPE LS-444/WIH AND TYPE LS-543/SIC

FIGURE 1 - MOUNTING DETAILS

Note: All dimensions shown are in Millimeters

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4. Finish. Metal surfaces shall be finished with a corrosive resistant latex or epoxy paint, in the specified color, in accordance with best commercial practices. Composite materials shall be finished in the color specified in the contract.

(b) Circuit label. A printed schematic of the loudspeaker enclosure wiring shall be permanently affixed to the inside of the enclosure where it will be easily readable with the cover open. It shall show wire color coding, terminal designations and selector switch connections and functions.

3. Reliability Speaker enclosures shall have a MTBF of 25,000 hours.

4. Types of loudspeakers. Loudspeakers shall be of the following types:

(a) Loudspeaker - LS-443/WIH
(b) Loudspeaker - LS-444/WIH
(c) Loudspeaker - LS-543()/SIC

5. Loudspeaker input. Loudspeakers built to this specification shall accept an input from a 70 7-volt line.

6 Switches.

(a) Channel selector. The channel selector shall be a 5-position switch. It shall be so connected that it will switch its two output leads to any one of 4 separate inputs. The unconnected switch position shall be at the counter clockwise (CCW) end of the rotation and shall be marked "OFF". The output of the channel selector shall be fed to the volume selector.

(b) Volume selector. The volume selector switch shall be a 6-position switch. It shall connect one lead from the channel selector to various taps of the line-to-voice coil transformer, with volume increasing in the clockwise (CW) direction. The other lead from the channel selector shall be permanently connected to the other side of the transformer.

7. Multiple speaker operation. To provide for multiple speaker operation, the two input leads to the volume selector switch shall be made available on a terminal strip.

8. Shock. The loudspeakers built to this specification shall meet the shock requirements of MIL-S-901 for Lightweight, Grade B, Class I, Type A, Hull mounted equipment.

9. Stray magnetic field. The maximum value of stray magnetic field measured on a horizontal plane 76 cm below the axis of the loudspeaker magnet shall not exceed 1.05 milligaus (105 gamma). The stray magnetic field limit shall apply only to LS-543()/SIC loudspeakers.

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Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices. The government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements. The offerer/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes.

1 Ordering data.

a. Title, number, and date of this Commercial Item Description.
b. Type of loudspeaker required.
c. Color desired

MILITARY INTERESTS:

Custodians
Army - ER
Navy - SH
Air Force - 99

Reviewing activity
Navy - AS

User activity
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITIES:
COMMERCE - NIST
GSA-FSS

PREPARING ACTIVITY:
NAVY - SH
(PROJECT 5965-0228)

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